

Thm. Let $f: [a, b] \times [c, d] \rightarrow \mathbb{R}$ be a continuous map. Define

$$\phi(x) = \int_c^d f(x, t) dt, \quad x \in [a, b].$$

Then, the following properties hold:

1) ϕ is continuous.

2) If the partial derivative $D_x f$ exists and continuous on $(a, b) \times [c, d]$, then ϕ is differentiable on (a, b) with the derivative is

$$\phi'(x) = \int_c^d D_x f(x, t) dt, \quad x \in (a, b).$$

Proof. Since $[a, b] \times [c, d]$ is compact, f is continuous uniformly on the domain. Therefore, given $\epsilon > 0$, there exists $\delta > 0$ such that:

$$|x - y| < \delta, \quad t \in [c, d] \text{ implies } |f(x, t) - f(y, t)| < \frac{\epsilon}{d - c}.$$

Now, by the property of integral, $|x - y| < \delta$ implies

$$|\phi(x) - \phi(y)| = \left| \int_c^d f(x, t) - f(y, t) dt \right| \leq \int_c^d |f(x, t) - f(y, t)| dt < (d - c) \cdot \frac{\epsilon}{d - c} = \epsilon.$$

So, the first assertion is proved. To show the second assertion, observe that

$$\int_x^{x+h} D_x f(z, t) dz = f(x+h, t) - f(x, t)$$

by the Fundamental Theorem of Calculus, since the $D_x f$ is continuous. Now,

$$\frac{1}{h} [\phi(x+h) - \phi(x)] = \frac{1}{h} \int_c^d f(x+h, t) - f(x, t) dt = \frac{1}{h} \int_c^d \int_x^{x+h} D_x f(z, t) dz dt$$

where $h \neq 0$, and $x \in (a, b)$ was arbitrary. Take $r > 0$ such that $x \in [x-r, x+r] \subseteq (a, b)$.

Then, $D_x f$ is continuous uniformly on $[x-r, x+r] \times [c, d]$, so that given $\epsilon > 0$, there exists $\delta > 0$ such that

$$|z - x| < \delta, \quad t \in [c, d] \text{ implies } |D_x f(z, t) - D_x f(x, t)| < \frac{\epsilon}{d - c}.$$

Finally,

$$\begin{aligned} \left| \frac{\phi(x+h) - \phi(x)}{h} - \int_c^d D_x f(x, t) dt \right| &= \left| \int_c^d \frac{1}{h} \cdot \int_x^{x+h} D_x f(z, t) - D_x f(x, t) dz dt \right| \\ &\leq \int_c^d \frac{1}{h} \cdot \int_x^{x+h} |D_x f(z, t) - D_x f(x, t)| dz dt < \epsilon. \quad \square \end{aligned}$$

